APPENDIX I. PAPUA NEW GUINEA: PUBLIC AND EXTERNAL DEBT SUSTAINABILITY[5]

This appendix summarizes the debt sustainability analysis (DSA) for Papua New Guinea using the new framework for low-income countries. The baseline macroeconomic scenario underlying this DSA is the medium-term outlook discussed in the staff report, including the sustained implementation of the ongoing medium-term fiscal strategy. This appendix also assesses the impact of various exogenous shocks on the sustainability of public sector debt and external debt under the baseline scenario.

A. Main Assumptions

- Real GDP growth is projected to grow at about 4 percent in 2006-2011 as agricultural productivity improves, several mining projects come on stream, and continued gradual progress is made on structural reform.
- Monetary policy continues to be geared toward maintaining low inflation and achieving a sound external position.
- Fiscal policy remains unchanged and in line with the government's medium-term fiscal strategy; surpluses are expected to decline due to falling mineral revenues and nonmineral balances improve steadily after 2007.
- Papua New Guinea would have access to external concessional finance and continue to follow a prudent debt strategy.
- Prices of key commodities are in line with January 2007 WEO assumptions.
- Structural reforms outlined under the authorities' Medium-term Development Strategy, including those on public sector reform, are implemented at a gradual pace.
- Given the considerable uncertainty surrounding the PNG-Queensland gas pipeline project, this DSA assumes no government borrowing to finance an equity participation (in contrast to that envisaged in the 2005 Article IV DSA).

[5] As Papua New Guinea is an IBRD/IDA blend country, this DSA was prepared by Fund staff only under the IMF-WB DSA framework for Low-Income Countries. See *Operational Framework for Debt Sustainability Assessments in Low-Income Countries—Further Considerations*.

B. Total Public Sector Debt Sustainability

The public sector debt ratio declined from 67.4 percent of GDP at end-2003 to an estimated 39.3 percent of GDP at end-2006 (Appendix I, Table 1). This improvement reflects prudent fiscal policy, favorable mineral sector developments (in particular, strong export prices in recent years), kina appreciation, lower interest rates, and a sustained recovery since the current government took office in mid-2002. Of total public debt, approximately 50 percent is external debt, of which 47 percent of loans are in US dollars, 36 percent in Japanese yen, and 14 percent in Euros. Multilateral debt accounts for about two-thirds (AsDB and World Bank) of external debt. Japan is the main bilateral creditor. Under the authorities' Medium-Term Debt Strategy, foreign-currency loans are considered only if they are highly concessionary (grant element of 35 percent).

Under the baseline, the public sector debt ratio is projected to decline to 34.3 percent of GDP by 2011 and 13.6 percent by 2026. The projected debt improvement stems mainly from the expectation of a continued prudent fiscal policy into the medium term, and favorable debt dynamics. The NPV of public debt would decline sharply from 48.2 percent of GDP in 2005 to 40.1 percent of GDP in 2006, and continue to decline to 35.3 percent of GDP by 2011 and to 13.5 percent of GDP by 2026. The NPV of public debt service-to-revenue ratio, which is estimated to fall significantly in 2006 due to high windfall revenue, is expected to rise somewhat in 2007 as mineral revenue moderates. The debt service to revenue ratio should revert to being relatively stable (13 percent), before declining sharply in the longer term.

Papua New Guinea faces a moderate risk of debt distress under the IMF-WB DSA framework for LICs.[6] The main results are: (i) only one indicator—NPV of debt to GDP ratio—is above its threshold under the baseline scenario—and it declines under the threshold over the long term, (ii) other indicators are well below their thresholds over the entire projected period, and (iii) only under one bound test for lower real GDP growth are these thresholds significantly breached.

The standard analysis indicates that the paths of the debt ratios are particularly sensitive to changes in real GDP growth and the exchange rate (Appendix I, Table 2 and Figure 1). Compared to 2006, a decline in real GDP growth to one standard deviation below its historical average would raise the NPV of public debt to GDP ratio by about 20 percentage points of GDP in the long term. A one-time 30 percent real exchange rate

[6] Papua New Guinea is rated a poor performer for its policies and institutions by the IMF-WB LIC DSA framework. Thus, Papua New Guinea's applicable thresholds are: (i) 100 percent for NPV of debt-to-exports, (ii) 30 percent for NPV of debt-to-GDP, and (iii) 200 percent for NPV of debt-to-fiscal revenue; and for debt service: (i) 15 percent of exports and (ii) 25 percent of revenues.

depreciation in 2007 would raise the NPV of debt to GDP ratio by about 10 percentage points.

C. External Debt Sustainability

Papua New Guinea's total external debt has declined in recent years, from 65.3 percent of GDP in 2003 to 32.9 percent in 2006 (Appendix I, Table 3). External debt as a percent of exports is also estimated to have declined significantly from 107 percent in 2003 to 42 percent in 2006.

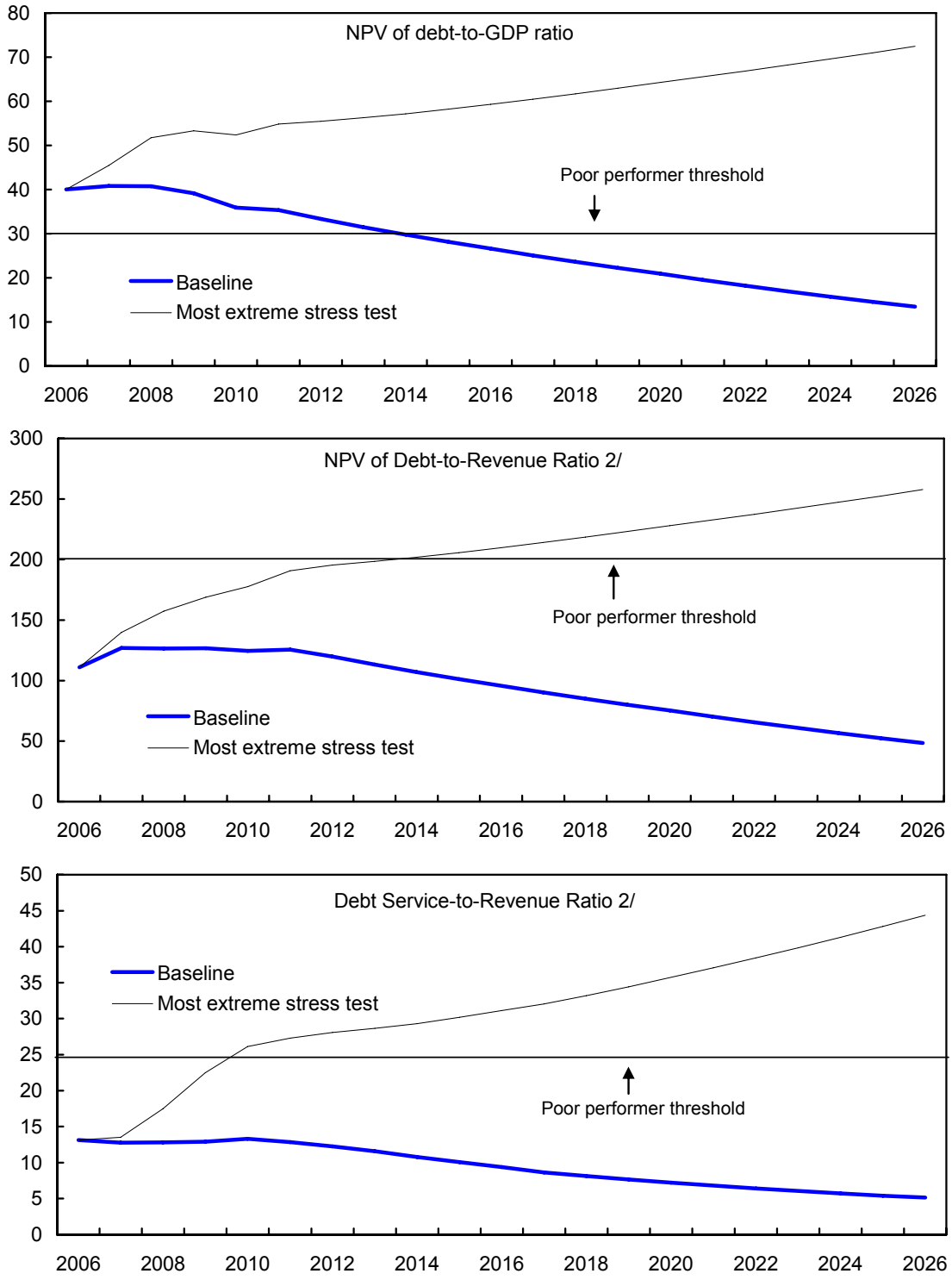
External public debt dynamics are sustainable under the baseline scenario, as all three debt indicators (the NPV of public external debt of GDP, the NPV of debt in percent of exports, and the debt service to exports ratio) do not breach the key thresholds. Based on available data, private external debt was estimated at 12 percent of GDP at end-2006. Private external debt is expected to decline over the long term, as large mineral projects that accounted for a high proportion of debt reach completion.

External debt sustainability is most vulnerable to shocks to lower export growth. Stress tests were carried out to test the robustness of the baseline scenario. The decline in export value growth by one standard deviation below its historical average would raise the NPV of public external debt-to-GDP threshold by 25 percentage points during 2006–09, and only fall below the threshold level in 2019. A one time 30 percent nominal depreciation in 2007 would raise the NPV of debt to GDP by 6 percentage points in the first three years before falling over the longer term. The export shock to the NPV of debt-to-exports would raise the level closer to (though still under) the distress threshold for the first ten years, before falling over the long term. Papua New Guinea's external debt distress risk is further moderated by the fact that it has an excellent external repayment record, having never defaulted in the past.

D. Conclusions

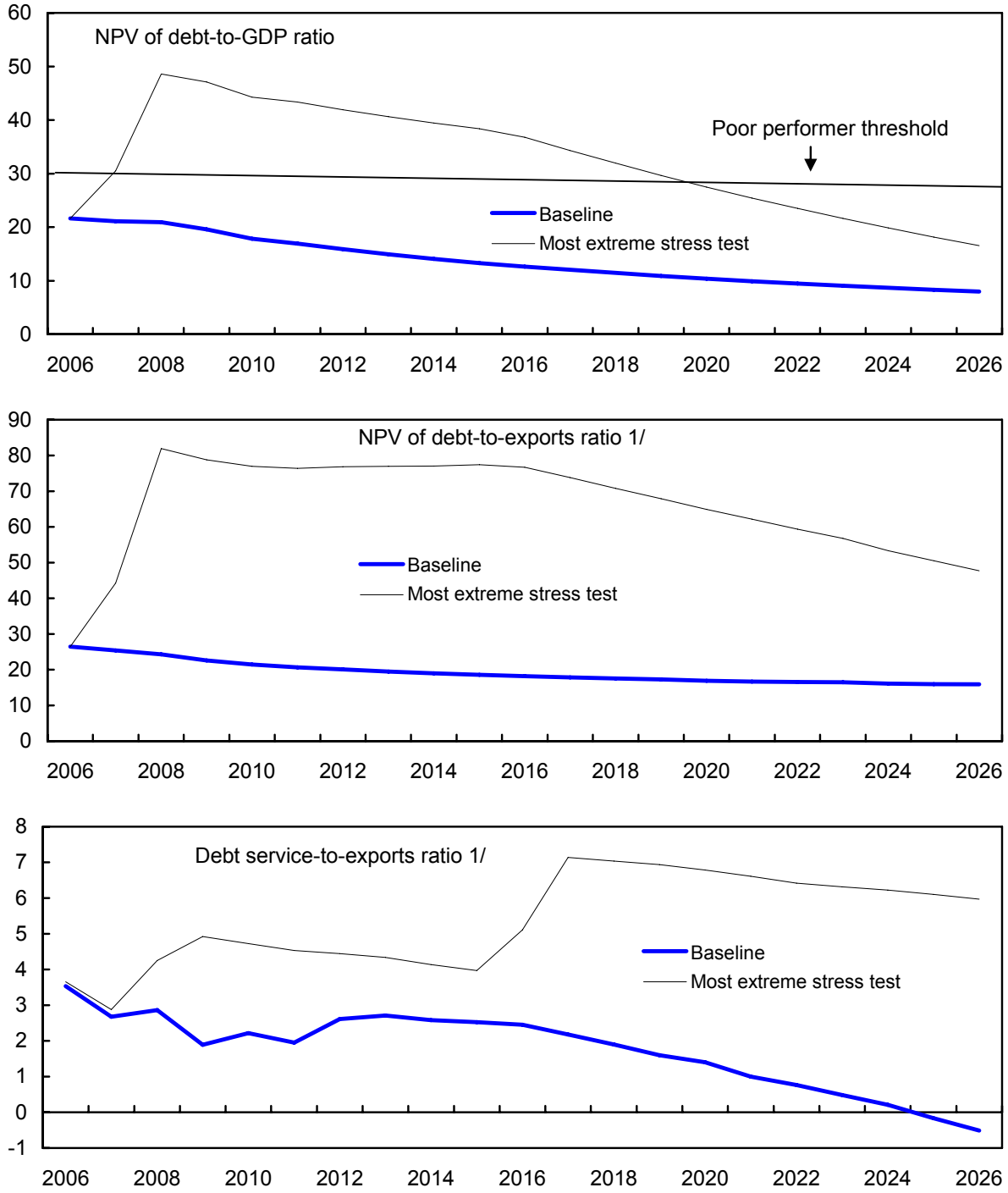
While Papua New Guinea has made significant progress in reducing its public debt burden, it will need to continue its prudent management of debt over the medium-to-long term in order to mitigate risks to public debt sustainability. While most indicators show that the economy's debt is manageable, Papua New Guinea faces a moderate risk of debt distress. The outlook is particularly sensitive to changes in real GDP growth and to the exchange rate, highlighting the vulnerability of the debt dynamics to potential shocks and the need to avoid policy slippages. To further help guard against these vulnerabilities, Papua New Guinea should ensure that any external borrowing is obtained on concessional terms, in line with its Medium-Term Debt Strategy.

Appendix I. Figure 1. Papua New Guinea: Indicators of Public Debt Under Alternative Scenarios, 2006-2026 1/



Sources: Papua New Guinea authorities; and Fund staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2016.
2/ Revenue including grants.

Appendix I. Figure 2. Papua New Guinea: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2006-2026
(In percent)



Sources: Papua New Guinea authorities; and Fund staff projections and simulations.

1/ Poor performer threshold for NPV of debt-to-exports ratio is 100 percent and for debt service-to-exports ratio is 15 percent.

Appendix I. Table 1. Papua New Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2003-2026

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2003	2004	2005	Historical Average 5/	Standard Deviation 5/	2006	2007	2008	2009	2010	2011	2006-11 Average	2016	2026	2012-26 Average
Public sector debt 1/	67.4	62.0	47.5			39.3	39.9	39.8	38.1	35.0	34.3		25.2	13.6	
o/w foreign-currency denominated	42.9	36.8	25.3			20.7	20.6	19.9	18.8	16.8	16.1		11.3	8.2	
Change in public sector debt	-9.2	-5.4	-14.5			-8.1	0.6	-0.1	-1.7	-3.2	-0.6		-1.6	-0.8	
Identified debt-creating flows	-12.8	-3.3	-2.6			-2.0	7.1	5.3	3.7	1.2	2.5		-0.8	-0.4	
Primary deficit	-3.7	-3.0	-6.3	-2.0	2.4	-9.0	-4.8	-4.9	-3.7	-3.0	-2.5	-4.6	-1.6	-0.9	-1.4
Revenue and grants	29.1	34.1	34.2			36.1	32.1	32.2	30.9	28.8	28.2		27.8	27.8	
of which : grants	5.9	6.7	8.0			5.8	5.0	5.1	5.0	4.6	4.4		3.4	2.1	
Primary (noninterest) expenditure	25.4	31.2	27.8			27.1	27.3	27.3	27.2	25.8	25.6		26.2	26.9	
Automatic debt dynamics	-8.8	-0.2	-8.9			-3.3	2.8	2.1	0.6	-1.1	0.8		0.0	0.0	
Contribution from interest rate/growth differential	1.8	0.6	-4.2			-1.3	1.1	1.2	0.0	-0.6	0.3		0.0	0.0	
of which : contribution from average real interest rate	3.4	2.4	-2.2			0.4	2.7	2.6	1.7	1.1	1.6		0.9	0.5	
of which : contribution from real GDP growth	-1.6	-1.8	-2.0			-1.7	-1.6	-1.4	-1.7	-1.7	-1.3		-0.9	-0.5	
Contribution from real exchange rate depreciation	-10.6	-0.8	-4.8			-2.0	1.7	1.0	0.6	-0.5	0.5		...	...	
Other identified debt-creating flows	-0.3	-0.2	12.7			10.3	9.1	8.1	6.7	5.3	4.2		0.8	0.4	
Privatization receipts (negative)	-0.3	-0.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	12.7			10.3	9.1	8.1	6.7	5.3	4.2		0.8	0.4	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	3.7	-2.0	-11.9			-6.2	-6.4	-5.5	-5.3	-4.4	-3.2		-0.8	-0.4	
NPV of public sector debt	24.5	25.2	48.2			40.1	40.8	40.8	39.2	35.9	35.3		26.6	13.5	
o/w foreign-currency denominated	0.0	0.0	26.1			21.5	21.5	20.9	19.9	17.8	17.2		12.8	8.1	
o/w external	...	...	26.1			21.5	21.5	20.9	19.9	17.8	17.2		12.8	8.1	
NPV of contingent liabilities (not included in public sector debt)	0.0	0.0	9.7			10.0	10.0	10.2	9.8	9.2	8.9		6.3	4.1	
Gross financing need 2/	23.5	24.5	10.9			4.6	3.0	-0.6	0.6	1.5	1.9		1.6	0.9	
NPV of public sector debt-to-revenue ratio (in percent) 3/	84.2	73.7	141.0			111.0	127.0	126.6	126.8	124.6	125.5		95.7	48.4	
o/w external	...	...	76.2			59.5	67.0	64.9	64.3	61.8	61.0		45.9	29.0	
Debt service-to-revenue ratio (in percent) 3/ 4/	34.6	18.1	13.2			13.2	12.8	12.8	12.9	13.3	12.8		9.4	5.1	
Primary deficit that stabilizes the debt-to-GDP ratio	5.4	2.4	8.2			-0.8	-5.4	-4.8	-2.0	0.2	-1.9		0.0	0.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	2.2	2.7	3.3	1.2	3.7	3.7	4.3	3.7	4.4	4.7	4.0	4.1	3.5	3.5	3.5
Average nominal interest rate on forex debt (in percent)	2.7	2.6	2.6	3.6	0.8	2.9	3.3	3.4	3.5	3.6	3.4	3.4	3.0	1.6	2.7
Average real interest rate on domestic currency debt (in percent)	13.9	10.0	-9.0	3.2	6.6	0.7	13.5	12.4	7.5	4.7	8.1	7.8	5.7	9.4	6.5
Real exchange rate depreciation (in percent, + indicates depreciation)	-19.8	-1.8	-13.4	3.3	13.6	-8.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	5.5	-2.0	17.3	8.4	5.7	8.6	-2.1	-2.7	1.6	4.9	1.5	2.0	3.5	3.5	3.5
Growth of real primary spending (deflated by GDP deflator, in percent)	-11.1	26.2	-7.7	3.8	15.1	1.1	5.1	3.6	4.0	-0.5	3.0	2.7	3.9	3.7	3.8
Grant element of new external borrowing (in percent)	...	...	...	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...

Sources: Papua New Guinea authorities; and Fund staff estimates and projections.

1/ Includes central government, central bank external debt, and statutory authorities.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Appendix I. Table 2. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public Debt 2006-2026

	Estimate	Projections						
	2006	2007	2008	2009	2010	2011	2016	2026
NPV of Debt-to-GDP Ratio								
Baseline	40	41	41	39	36	35	27	13
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	40	45	49	51	50	51	47	34
A2. Primary balance is unchanged from 2006	40	37	33	26	18	11	-29	-106
A3. Permanently lower GDP growth 1/	40	41	42	41	39	40	39	57
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007-2008	40	45	52	53	52	55	59	72
B2. Primary balance is at historical average minus one standard deviations in 2007-2008	40	46	51	49	45	45	36	22
B3. Combination of B1-B2 using one half standard deviation shocks	40	47	53	51	47	47	36	19
B4. One-time 30 percent real depreciation in 2007	40	51	52	50	46	45	36	20
B5. 10 percent of GDP increase in other debt-creating flows in 2007	40	50	50	49	45	45	35	21
NPV of Debt-to-Revenue Ratio 2/								
Baseline	111	127	127	127	125	126	96	48
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	111	139	151	162	169	178	164	116
A2. Primary balance is unchanged from 2006	111	115	102	85	63	40	-104	-383
A3. Permanently lower GDP growth 1/	111	128	130	133	135	140	139	203
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007-2008	111	140	157	169	178	191	210	258
B2. Primary balance is at historical average minus one standard deviations in 2007-2008	111	142	158	159	158	160	129	78
B3. Combination of B1-B2 using one half standard deviation shocks	111	145	163	164	161	163	127	69
B4. One-time 30 percent real depreciation in 2007	111	160	160	162	160	162	128	71
B5. 10 percent of GDP increase in other debt-creating flows in 2007	111	156	156	158	156	158	127	77
Debt Service-to-Revenue Ratio 2/								
Baseline	13	13	13	13	13	13	9	5
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	13	13	18	22	24	23	19	15
A2. Primary balance is unchanged from 2006	13	13	6	2	-3	-8	-32	-77
A3. Permanently lower GDP growth 1/	13	13	13	14	15	16	18	35
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007-2008	13	14	17	22	26	27	31	44
B2. Primary balance is at historical average minus one standard deviations in 2007-2008	13	13	21	27	22	19	15	11
B3. Combination of B1-B2 using one half standard deviation shocks	13	13	21	26	22	18	14	9
B4. One-time 30 percent real depreciation in 2007	13	13	14	15	15	15	11	7
B5. 10 percent of GDP increase in other debt-creating flows in 2007	13	13	29	23	20	17	15	11

Sources: Papua New Guinea authorities; and Fund staff projections and simulations.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Appendix I. Table 3. Papua New Guinea: External Debt Sustainability Framework, Baseline Scenario, 2003-2026 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2003	2004	2005			Estimate 2006	2007	2008	2009	2010	2011	2006-11 Average	2016	2026	2012-26 Average
External debt (nominal) 1/	**65.3**	**55.0**	**40.5**			**32.9**	**30.8**	**29.3**	**25.7**	**22.2**	**19.8**	**26.8**	**14.8**	**11.5**	
o/w public and publicly guaranteed (PPG)	45.7	38.0	25.4			20.9	20.2	19.9	18.6	16.8	15.9	**18.7**	11.2	8.1	
Change in external debt	-11.5	-10.4	-14.5			-7.6	-2.1	-1.4	-3.6	-3.6	-2.4		-0.8	-0.1	
Identified net debt-creating flows	-18.9	-9.4	-16.5			-11.0	-5.7	-4.8	-4.1	-3.1	-2.6		0.6	0.7	
Non-interest current account deficit	**-7.1**	**-4.0**	**-5.2**	**-5.5**	**4.0**	**-10.0**	**-5.0**	**-4.3**	**-3.4**	**-3.0**	**-2.6**	**-4.7**	**0.5**	**0.7**	0.3
Deficit in balance of goods and services	-2.0	1.6	6.9			-2.9	1.8	3.9	3.8	4.6	5.3		8.6	5.9	
Exports	67.4	70.2	71.0			81.8	83.0	85.9	86.6	83.2	82.1		69.4	50.3	
Imports	65.4	71.8	77.9			79.0	84.8	89.8	90.4	87.8	87.4		78.0	56.2	
Net current transfers (negative = inflow)	-6.6	-4.6	-8.8	-5.5	1.6	-6.2	-5.4	-5.7	-5.6	-5.1	-4.9		-3.8	-2.2	-3.3
Other current account flows (negative = net inflow)	1.5	-0.9	-3.3			-0.9	-1.4	-2.4	-1.6	-2.5	-3.0		-4.3	-3.0	
Net FDI (negative = inflow)	**-2.7**	**-0.7**	**-1.4**	**-2.8**	**2.5**	**-2.3**	**-1.9**	**-2.0**	**-1.9**	**-1.3**	**-1.3**	**-1.8**	**-1.0**	**-0.6**	-0.9
Endogenous debt dynamics 2/	**-9.0**	**-4.8**	**-10.0**			**1.3**	**1.2**	**1.5**	**1.2**	**1.2**	**1.3**	**1.3**	**1.1**	**0.6**	
Contribution from nominal interest rate	2.7	1.7	1.3			2.6	2.6	2.6	2.5	2.3	2.2		1.6	1.0	
Contribution from real GDP growth	-1.4	-1.6	-1.5			-1.3	-1.4	-1.1	-1.2	-1.1	-0.8		-0.5	-0.4	
Contribution from price and exchange rate changes	-10.3	-4.9	-9.8			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**7.4**	**-1.0**	**2.1**			**3.4**	**3.6**	**3.3**	**0.5**	**-0.5**	**0.2**	**1.8**	**-1.4**	**-0.8**	
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt 4/	…	…	41.2			33.6	31.6	30.3	26.7	23.1	20.8		16.3	11.3	
In percent of exports	…	…	58.1			41.1	38.1	35.3	30.9	27.8	25.3		23.4	22.6	
NPV of PPG external debt	**…**	**…**	**26.1**			**21.6**	**21.1**	**20.9**	**19.6**	**17.8**	**16.9**	**19.7**	**12.6**	**8.0**	
In percent of exports	…	…	36.8			26.4	25.4	24.3	22.6	21.4	20.6		18.2	15.9	
Debt service-to-exports ratio (in percent)	16.2	12.7	9.9			8.3	7.0	7.1	6.7	6.5	6.3		5.7	4.3	
PPG debt service-to-exports ratio (in percent)	**0.8**	**6.2**	**4.9**			**3.5**	**2.7**	**2.9**	**1.9**	**2.2**	**2.0**	**2.5**	**2.5**	**-0.5**	
Total gross financing need (billions of U.S. dollars)	0.0	0.2	0.0			-0.3	-0.1	0.0	0.0	0.1	0.1		0.3	0.3	
Non-interest current account deficit that stabilizes debt ratio	4.3	6.4	9.3			-2.4	-2.9	-2.8	0.2	0.6	-0.2		1.4	0.8	
Key macroeconomic assumptions															
Real GDP growth (in percent)	2.2	2.7	3.3	1.2	3.7	3.7	4.3	3.7	4.4	4.7	4.0	4.1	3.5	3.5	3.5
GDP deflator in US dollar terms (change in percent)	15.4	8.1	21.8	0.4	14.1	10.3	-3.2	-4.7	0.3	3.5	0.2	1.1	1.9	1.9	1.9
Effective interest rate (percent) 5/	4.1	2.9	3.0	4.2	0.7	7.3	7.9	8.4	8.8	9.7	10.1	8.7	10.9	8.7	10.5
Growth of exports of G&S (US dollar terms, in percent)	31.8	15.8	27.3	3.1	18.1	31.8	2.3	2.3	5.6	4.1	2.8	8.1	2.1	1.9	2.1
Growth of imports of G&S (US dollar terms, in percent)	13.6	21.9	36.6	7.3	17.5	15.9	8.3	4.7	5.5	5.3	3.6	7.2	2.5	2.1	2.5
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	35.3	39.7	37.9	36.8	39.0	33.1	37.0	32.5	32.6	32.6
Memorandym item:															
Nominal GDP (billions of US dollars)	3.5	3.9	4.9			5.7	5.7	5.6	5.9	6.4	6.7		8.7	14.9	

Sources: Papua New Guinea authorities; and Fund staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Appendix I. Table 4. Papua New Guinea: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2006-26
(In percent)

	Estimate	Projections						
	2006	2007	2008	2009	2010	2011	**2016**	2026
NPV of debt-to-GDP ratio								
Baseline	22	21	21	20	18	17	**13**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	22	20	17	14	11	7	**-20**	-73
A2. New public sector loans on less favorable terms in 2007-26 2/	22	21	21	20	19	18	**15**	12
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	22	23	24	22	20	19	**14**	9
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	22	31	49	47	44	43	**37**	17
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	22	24	26	24	22	21	**16**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08	22	23	26	24	22	21	**17**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	22	29	42	41	38	37	**30**	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	22	31	30	28	26	25	**18**	12
NPV of debt-to-exports ratio								
Baseline	26	25	24	23	21	21	**18**	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	26	24	20	17	13	9	**-28**	-146
A2. New public sector loans on less favorable terms in 2007-26 2/	26	26	25	24	23	22	**22**	23
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	26	25	24	23	21	21	**18**	16
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	26	44	82	79	77	76	**77**	48
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	26	25	24	23	21	21	**18**	16
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08	26	28	30	28	27	26	**24**	19
B5. Combination of B1-B4 using one-half standard deviation shocks	26	35	50	48	47	46	**45**	30
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	26	25	24	23	21	21	**18**	16
Debt service ratio								
Baseline	4	3	3	2	2	2	**2**	-1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	4	2	2	2	2	2	**1**	-8
A2. New public sector loans on less favorable terms in 2007-26 2/	4	2	3	3	2	2	**2**	2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	4	2	3	3	2	2	**2**	1
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	4	3	4	5	5	5	**5**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	4	2	3	3	2	2	**2**	1
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08	4	2	3	3	3	2	**2**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	4	3	3	4	3	3	**3**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	4	2	3	3	2	2	**2**	1
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	33	33	33	33	33	33	**33**	33

Sources: Papua New Guinea authorities; and Fund staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (in
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.